

JA3 & JA4+



A "Secret" Fingerprint
Identifying Bots and Scrapers

whoami

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Clients and Servers

- Clients want to stay anonymous
- Servers want to identify clients
- Both sides have merit
- Today -> Cat and Mouse Game

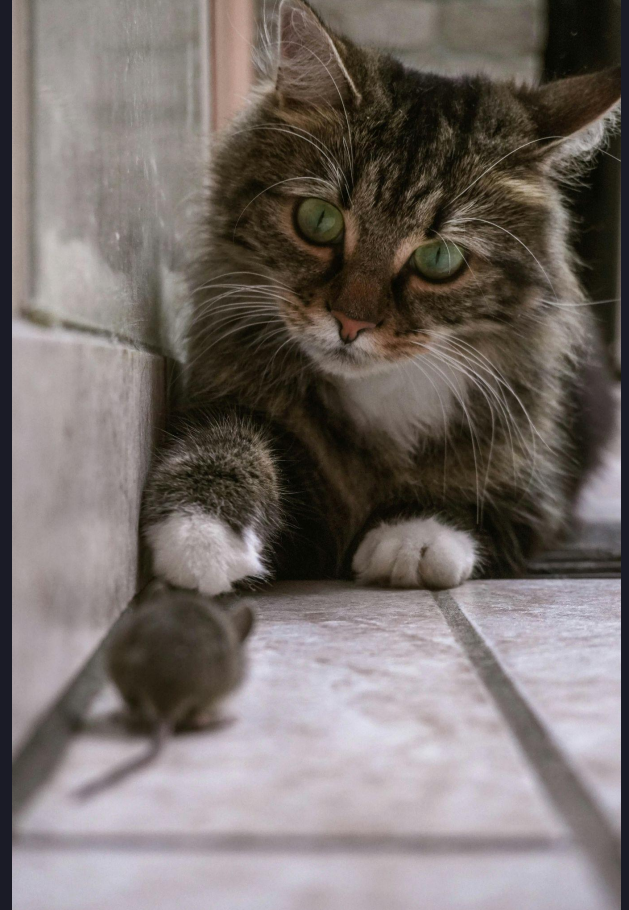
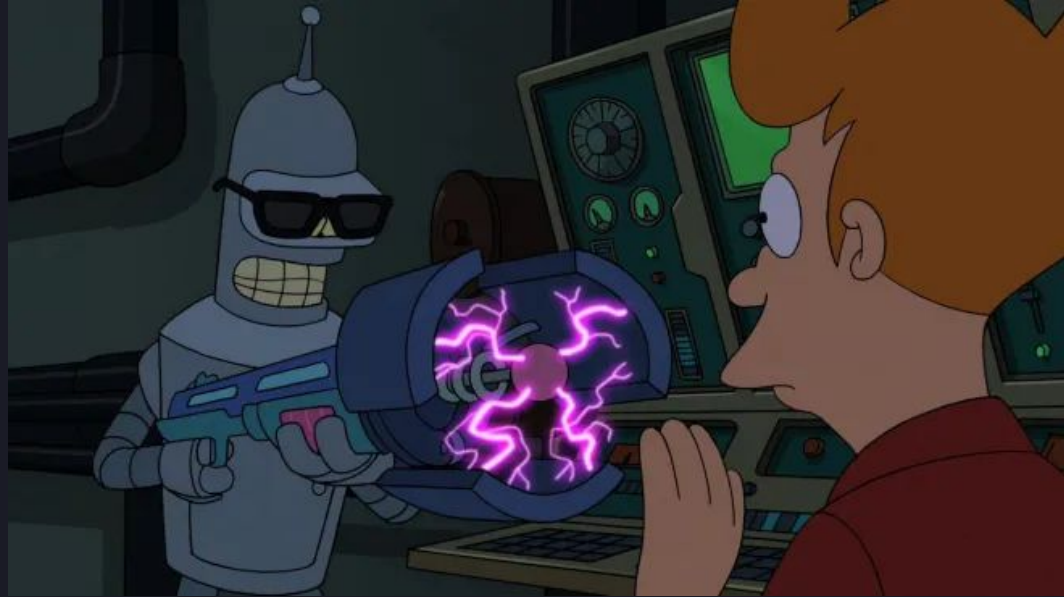


Photo by Katherine Mihailova

Robots On The Internet?

- Automated Attacks
 - Especially WordPress
- Scraping
- Plain Load - DDoS



Credit to Futurama (pls no sue)

JA3

TLS Fingerprinting Timeline

2015 - TLS Fingerprint idea ([blog](#))

2017 - ja3 ([github](#))

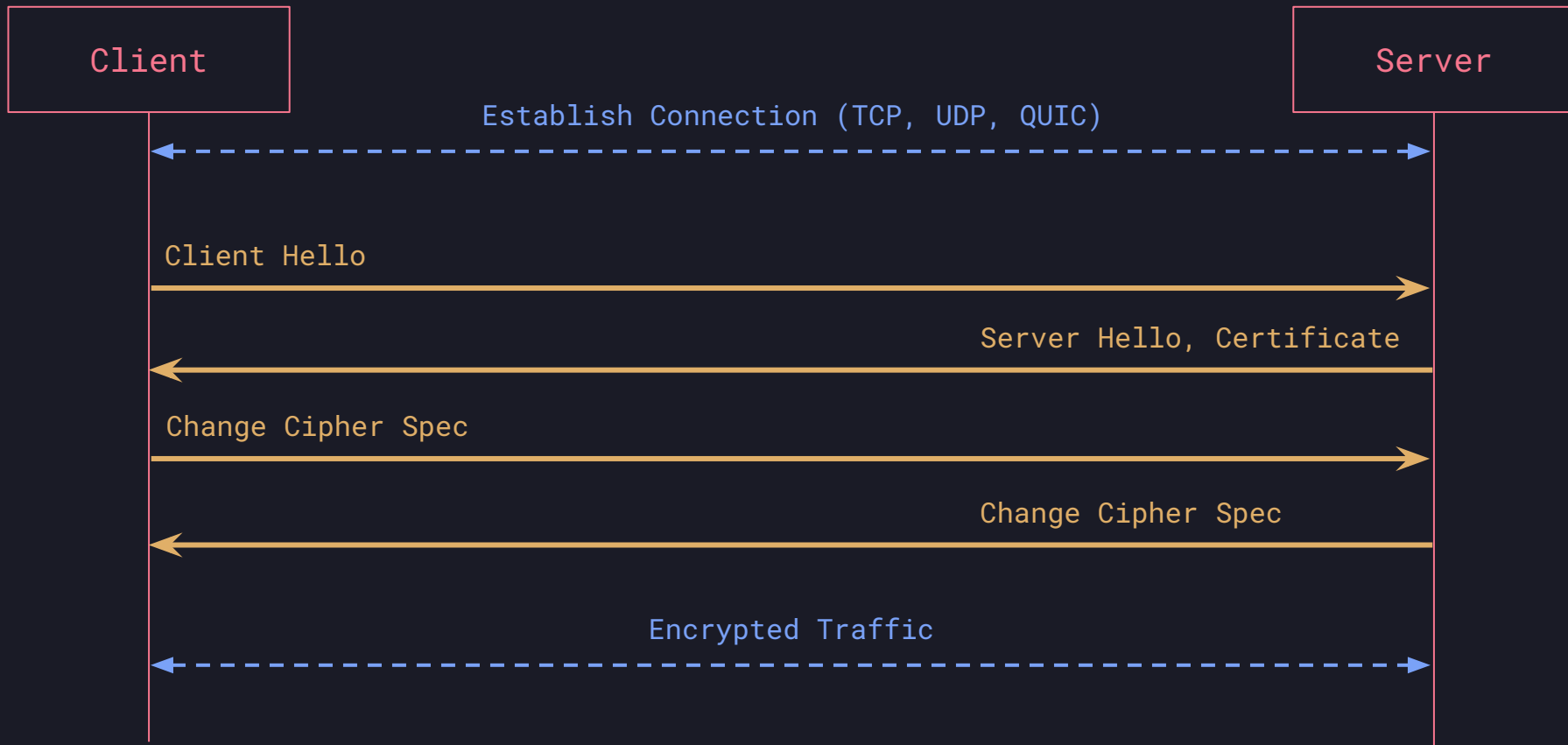
JA3

- John Althouse
- Jeff Atkinson
- Josh Atkins
- Thanks Salesforce!



<https://engineering.salesforce.com/tls-fingerprinting-with-ja3-and-ja3s-247362855967/>

SSL/TLS Client Hello



Parsing PCAPs

- 40+ years of effort
- ~10000 RFCs
- Good luck.

```
0000 f0 2f 74 09 12 c0 64 6e e0 90 43 45 08 00 45 00 /t...dn...CE...e
0010 02 c7 f8 38 40 00 40 06 3d 89 c0 a8 32 67 a4 44 ...8@...@...=...2g.D
0020 6b 1b c7 88 01 bb 03 d5 1b 8f 3a 1d c5 bb 80 18 k...:...
0030 01 f6 05 29 00 00 01 01 08 0a 03 73 19 80 8b 7f ...)....s....
0040 f4 bb 16 03 01 02 8e 01 00 02 8a 03 03 ef b4 68 .....h
0050 23 a9 6c ae 35 89 23 6c 51 0d 68 eb f7 02 c3 8a #.l.5.#l Q.h....
0060 0e 86 23 ba 4d ac 4a b9 cd 58 1d fc 9b 20 49 fd ..#.M.J...X...I.
0070 65 88 31 9d 4b 80 09 2c 98 e9 f6 5e 49 40 42 05 e.1.K..., ...^I@B.
0080 8f 9d 23 46 00 4d f9 43 fc c9 21 3e 3a d5 00 22 ..#F.M.C...!>:...
0090 13 01 13 03 13 02 c0 2b c0 2f cc a9 cc a8 c0 2c .....+.../.....,
00a0 c0 30 c0 0a c0 09 c0 13 c0 14 00 9c 00 9d c0 2f .0...../
00b0 00 35 01 00 02 1f 00 00 00 1a 00 18 00 00 15 62 .5.....b
00c0 6c 6f 67 2e 6d 69 6c 6f 73 6c 61 76 68 6f 6d 65 log.milo slavhome
00d0 72 2e 63 7a 00 17 00 00 ff 01 00 01 00 00 0a 00 r.cz....
00e0 0e 00 0c 00 1d 00 17 00 18 00 19 01 00 01 01 00 18 00 19 01 00 01 01 00
00f0 0b 00 02 01 00 00 10 00 0e 00 0c 02 68 32 08 68 .....h2.h
0100 74 74 70 2f 31 2e 31 00 05 00 05 01 00 00 00 00 ttp/1.1. ....
0110 00 22 00 0a 00 08 04 03 05 03 06 03 02 03 00 33 .".....3
0120 00 6b 00 69 00 1d 00 20 d6 04 83 22 00 0f ce 42 .k.i...."....B
0130 15 53 70 75 9d ca 18 f7 ce a5 6e e0 5d 01 06 2a .Spu....n.]...*
0140 c8 14 0b f0 f1 91 55 51 00 17 00 41 04 21 a8 db .....UQ...A.!..
0150 b1 41 a4 ae 93 b0 09 ba 27 06 3f 7a 1c ff 9a 80 .A.....'?z....
0160 d0 5a cf ad 8b 00 3c 98 7b 3f a6 00 43 7c db 2e .Z....< {?.C|.
0170 19 b2 74 ad 5c 0f 75 2f c9 8c d1 08 6a 0a 11 06 ..t.\.u/ ...j...
0180 0a b3 95 45 5a 87 57 e8 d9 61 c5 6b c7 00 2b 00 ...EZ.W...a.k...+
0190 05 04 03 04 03 03 00 0d 00 18 00 16 04 00 05 03 .. ..?z....
01a0 06 03 08 04 08 05 08 06 04 01 05 01 06 01 02 03 .....
01b0 02 01 00 1c 00 02 40 01 fe 0d 01 19 00 00 01 00 .....@...
01c0 03 ab 00 20 f9 9d 62 53 29 e0 1a 6a ff c4 3e 75 ... ..bS )..j...>u
01d0 81 38 d3 dd 7c ff 50 b7 f5 b6 8e 11 b7 28 03 f1 .8...|.P.....(
01e0 af a6 15 18 00 ef 98 e7 64 ac fb de 0a a9 98 e2 .....d.....
01f0 91 7a 86 4c 01 77 fb f4 ae f4 3c a7 ce df 93 c4 .z.L.w...<....
0200 e0 9c 40 b6 3a 59 0f 00 b0 0a 6e 69 6c df 0b 79 ..@.:Y...nil..y
0210 5c 69 cb 99 65 92 79 d9 53 d8 b0 a3 31 ac 68 13 \i..e.Y S...1.h
0220 d8 0e c7 0c 96 6d 41 9c 80 f0 8f 60 79 8b 85 6a .....mA...y..j
0230 75 bb 1c 51 69 b8 cf 15 7a e1 54 60 bd ae b5 37 u..Qi...z.T...7
0240 55 42 26 58 04 e6 94 81 88 49 01 75 a4 43 d6 22 UB&X....I.u.C..
0250 dc 9d 5e 79 4f 60 6f d0 f9 7e b0 c1 ee 6c 3f a9 ..^yo`o...~...l?
0260 e8 d8 4f 56 2b 2f de b2 41 a7 c9 b2 15 a0 8b 82 ..OV+/. A.....
0270 04 83 6b 0f f7 59 9b 67 c9 04 1d 8e 41 70 36 15 ...k..Y.g ...Ap6
0280 03 96 02 2b de 83 bf 43 4d f1 87 e6 6e 00 3c d1 ..+...C M...n.<
0290 52 00 3f 5b 42 05 fa bf 83 5f 24 b5 6a fc fe 66 R.?[B..._$.j..f
02a0 b8 04 60 cc 69 cc f0 85 25 a7 85 5b 16 e3 83 3a ..i...%..[...
02b0 dc 58 ad 9d 7f 6d 0a b6 25 81 5d a6 19 5b 9b 39 .X...m...%..[.9
02c0 f5 dc 3a 82 f9 1f eb cf 8b af b2 c6 27 4e c7 f1 ..:.....'N..
02d0 83 43 c5 3e 2b .C.>+
```

```

> Frame 272: Packet, 725 bytes on wire (5800 bits), 725 bytes captured (5800 bits) on interface wlp0s20f3, id 0
> Ethernet II, Src: Intel_90:43:45 (64:6e:e0:90:43:45), Dst: ASUSTekCOMPU_09:12:c0 (f0:2f:74:09:12:c0)
> Internet Protocol Version 4, Src: 192.168.50.103, Dst: 164.68.107.27
> Transmission Control Protocol, Src Port: 51080, Dst Port: 443, Seq: 1, Ack: 1, Len: 659
- Transport Layer Security
  [Stream index: 8]
  - TLSv1.3 Record Layer: Handshake Protocol: Client Hello
    Content Type: Handshake (22)
    Version: TLS 1.0 (0x0301)
    Length: 654
    - Handshake Protocol: Client Hello
      Handshake Type: Client Hello (1)
      Length: 650
      - Version: TLS 1.2 (0x0303)
        [Expert Info (Chat/Deprecated): This legacy_version field MUST be ignored. The supported_versions extension is present and MUST be used instead.]
        [This legacy_version field MUST be ignored. The supported_versions extension is present and MUST be used instead.]
        [Severity level: Chat]
        [Group: Deprecated]
        Random: efb46823a96cae3589236c510d68ebf702c38a0e8623ba4dac4ab9cd581dfc9b
        Session ID Length: 32
        Session ID: 49fd6588319d4b80092c98e9f65e494042058f9d2346004df943fcc9213e3ad5
        Cipher Suites Length: 34
        Cipher Suites (17 suites)
        Compression Methods Length: 1
        Compression Methods (1 method)
        Extensions Length: 543
        Extension: server_name (len=26) name=blog.miloslavhomer.cz
        Extension: extended_master_secret (len=0)
        Extension: renegotiation_info (len=1)
        Extension: supported_groups (len=14)
        Extension: ec_point_formats (len=2)
        Extension: application_layer_protocol_negotiation (len=14)
        Extension: status_request (len=5)
        Extension: delegated_credentials (len=10)
        Extension: key_share (len=107) x25519, secp256r1
        - Extension: supported_versions (len=5) TLS 1.3, TLS 1.2
          Type: supported_versions (43)
          Length: 5
          Supported Versions length: 4
          Supported Version: TLS 1.3 (0x0304)
          Supported Version: TLS 1.2 (0x0303)
        Extension: signature_algorithms (len=24)
        Extension: record_size_limit (len=2)
        Extension: encrypted_client_hello (len=281)
        [JA4: t13d1713h2_5b57614c22b0_748f4c70de1c]
        [JA4_r: t13d1713h2_002f,0035,009c,009d,1301,1302,1303,c009,c00a,c013,c014,c02b,c02c,c02f,c030,cca8,cca9_0005,000a,000b,000d,0017,001c,0022,002b,0033,0034,0035,0036,0037,0038,0039,003a,003b,003c,003d,003e,003f,0040,0041,0042,0043,0044,0045,0046,0047,0048,0049,004a,004b,004c,004d,004e,004f,0050,0051,0052,0053,0054,0055,0056,0057,0058,0059,005a,005b,005c,005d,005e,005f,0060,0061,0062,0063,0064,0065,0066,0067,0068,0069,006a,006b,006c,006d,006e,006f,0070,0071,0072,0073,0074,0075,0076,0077,0078,0079,007a,007b,007c,007d,007e,007f,0080,0081,0082,0083,0084,0085,0086,0087,0088,0089,008a,008b,008c,008d,008e,008f,0090,0091,0092,0093,0094,0095,0096,0097,0098,0099,009a,009b,009c,009d,009e,009f,00a0,00a1,00a2,00a3,00a4,00a5,00a6,00a7,00a8,00a9,00aa,00ab,00ac,00ad,00ae,00af,00b0,00b1,00b2,00b3,00b4,00b5,00b6,00b7,00b8,00b9,00ba,00bb,00bc,00bd,00be,00bf,00c0,00c1,00c2,00c3,00c4,00c5,00c6,00c7,00c8,00c9,00ca,00cb,00cc,00cd,00ce,00cf,00d0,00d1,00d2,00d3,00d4,00d5,00d6,00d7,00d8,00d9,00da,00db,00dc,00dd,00de,00df,00e0,00e1,00e2,00e3,00e4,00e5,00e6,00e7,00e8,00e9,00ea,00eb,00ec,00ed,00ee,00ef,00f0,00f1,00f2,00f3,00f4,00f5,00f6,00f7,00f8,00f9,00fa,00fb,00fc,00fd,00fe,00ff,0100,0101,0102,0103,0104,0105,0106,0107,0108,0109,010a,010b,010c,010d,010e,010f,0110,0111,0112,0113,0114,0115,0116,0117,0118,0119,011a,011b,011c,011d,011e,011f,0120,0121,0122,0123,0124,0125,0126,0127,0128,0129,012a,012b,012c,012d,012e,012f,0130,0131,0132,0133,0134,0135,0136,0137,0138,0139,013a,013b,013c,013d,013e,013f,0140,0141,0142,0143,0144,0145,0146,0147,0148,0149,014a,014b,014c,014d,014e,014f,0150,0151,0152,0153,0154,0155,0156,0157,0158,0159,015a,015b,015c,015d,015e,015f,0160,0161,0162,0163,0164,0165,0166,0167,0168,0169,016a,016b,016c,016d,016e,016f,0170,0171,0172,0173,0174,0175,0176,0177,0178,0179,017a,017b,017c,017d,017e,017f,0180,0181,0182,0183,0184,0185,0186,0187,0188,0189,018a,018b,018c,018d,018e,018f,0190,0191,0192,0193,0194,0195,0196,0197,0198,0199,019a,019b,019c,019d,019e,019f,01a0,01a1,01a2,01a3,01a4,01a5,01a6,01a7,01a8,01a9,01aa,01ab,01ac,01ad,01ae,01af,01b0,01b1,01b2,01b3,01b4,01b5,01b6,01b7,01b8,01b9,01ba,01bb,01bc,01bd,01be,01bf,01c0,01c1,01c2,01c3,01c4,01c5,01c6,01c7,01c8,01c9,01ca,01cb,01cc,01cd,01ce,01cf,01d0,01d1,01d2,01d3,01d4,01d5,01d6,01d7,01d8,01d9,01da,01db,01dc,01dd,01de,01df,01e0,01e1,01e2,01e3,01e4,01e5,01e6,01e7,01e8,01e9,01ea,01eb,01ec,01ed,01ee,01ef,01f0,01f1,01f2,01f3,01f4,01f5,01f6,01f7,01f8,01f9,01fa,01fb,01fc,01fd,01fe,01ff,0200,0201,0202,0203,0204,0205,0206,0207,0208,0209,020a,020b,020c,020d,020e,020f,0210,0211,0212,0213,0214,0215,0216,0217,0218,0219,021a,021b,021c,021d,021e,021f,0220,0221,0222,0223,0224,0225,0226,0227,0228,0229,022a,022b,022c,022d,022e,022f,0230,0231,0232,0233,0234,0235,0236,0237,0238,0239,023a,023b,023c,023d,023e,023f,0240,0241,0242,0243,0244,0245,0246,0247,0248,0249,024a,024b,024c,024d,024e,024f,0250,0251,0252,0253,0254,0255,0256,0257,0258,0259,025a,025b,025c,025d,025e,025f,0260,0261,0262,0263,0264,0265,0266,0267,0268,0269,026a,026b,026c,026d,026e,026f,0270,0271,0272,0273,0274,0275,0276,0277,0278,0279,027a,027b,027c,027d,027e,027f,0280,0281,0282,0283,0284,0285,0286,0287,0288,0289,028a,028b,028c,028d,028e,028f,0290,0291,0292,0293,0294,0295,0296,0297,0298,0299,029a,029b,029c,029d,029e,029f,02a0,02a1,02a2,02a3,02a4,02a5,02a6,02a7,02a8,02a9,02aa,02ab,02ac,02ad,02ae,02af,02b0,02b1,02b2,02b3,02b4,02b5,02b6,02b7,02b8,02b9,02ba,02bb,02bc,02bd,02be,02bf,02c0,02c1,02c2,02c3,02c4,02c5,02c6,02c7,02c8,02c9,02ca,02cb,02cc,02cd,02ce,02cf,02d0,02d1,02d2,02d3,02d4,02d5,02d6,02d7,02d8,02d9,02da,02db,02dc,02dd,02de,02df,02e0,02e1,02e2,02e3,02e4,02e5,02e6,02e7,02e8,02e9,02ea,02eb,02ec,02ed,02ee,02ef,02f0,02f1,02f2,02f3,02f4,02f5,02f6,02f7,02f8,02f9,02fa,02fb,02fc,02fd,02fe,02ff,0300,0301,0302,0303,0304,0305,0306,0307,0308,0309,030a,030b,030c,030d,030e,030f,0310,0311,0312,0313,0314,0315,0316,0317,0318,0319,031a,031b,031c,031d,031e,031f,0320,0321,0322,0323,0324,0325,0326,0327,0328,0329,032a,032b,032c,032d,032e,032f,0330,0331,0332,0333,0334,0335,0336,0337,0338,0339,033a,033b,033c,033d,033e,033f,0340,0341,0342,0343,0344,0345,0346,0347,0348,0349,034a,034b,034c,034d,034e,034f,0350,0351,0352,0353,0354,0355,0356,0357,0358,0359,035a,035b,035c,035d,035e,035f,0360,0361,0362,0363,0364,0365,0366,0367,0368,0369,036a,036b,036c,036d,036e,036f,0370,0371,0372,0373,0374,0375,0376,0377,0378,0379,037a,037b,037c,037d,037e,037f,0380,0381,0382,0383,0384,0385,0386,0387,0388,0389,038a,038b,038c,038d,038e,038f,0390,0391,0392,0393,0394,0395,0396,0397,0398,0399,039a,039b,039c,039d,039e,039f,03a0,03a1,03a2,03a3,03a4,03a5,03a6,03a7,03a8,03a9,03aa,03ab,03ac,03ad,03ae,03af,03b0,03b1,03b2,03b3,03b4,03b5,03b6,03b7,03b8,03b9,03ba,03bb,03bc,03bd,03be,03bf,03c0,03c1,03c2,03c3,03c4,03c5,03c6,03c7,03c8,03c9,03ca,03cb,03cc,03cd,03ce,03cf,03d0,03d1,03d2,03d3,03d4,03d5,03d6,03d7,03d8,03d9,03da,03db,03dc,03dd,03de,03df,03e0,03e1,03e2,03e3,03e4,03e5,03e6,03e7,03e8,03e9,03ea,03eb,03ec,03ed,03ee,03ef,03f0,03f1,03f2,03f3,03f4,03f5,03f6,03f7,03f8,03f9,03fa,03fb,03fc,03fd,03fe,03ff,0400,0401,0402,0403,0404,0405,0406,0407,0408,0409,040a,040b,040c,040d,040e,040f,0410,0411,0412,0413,0414,0415,0416,0417,0418,0419,041a,041b,041c,041d,041e,041f,0420,0421,0422,0423,0424,0425,0426,0427,0428,0429,042a,042b,042c,042d,042e,042f,0430,0431,0432,0433,0434,0435,0436,0437,0438,0439,043a,043b,043c,043d,043e,043f,0440,0441,0442,0443,0444,0445,0446,0447,0448,0449,044a,044b,044c,044d,044e,044f,0450,0451,0452,0453,0454,0455,0456,0457,0458,0459,045a,045b,045c,045d,045e,045f,0460,0461,0462,0463,0464,0465,0466,0467,0468,0469,046a,046b,046c,046d,046e,046f,0470,0471,0472,0473,0474,0475,0476,0477,0478,0479,047a,047b,047c,047d,047e,047f,0480,0481,0482,0483,0484,0485,0486,0487,0488,0489,048a,048b,048c,048d,048e,048f,0490,0491,0492,0493,0494,0495,0496,0497,0498,0499,049a,049b,049c,049d,049e,049f,04a0,04a1,04a2,04a3,04a4,04a5,04a6,04a7,04a8,04a9,04aa,04ab,04ac,04ad,04ae,04af,04b0,04b1,04b2,04b3,04b4,04b5,04b6,04b7,04b8,04b9,04ba,04bb,04bc,04bd,04be,04bf,04c0,04c1,04c2,04c3,04c4,04c5,04c6,04c7,04c8,04c9,04ca,04cb,04cc,04cd,04ce,04cf,04d0,04d1,04d2,04d3,04d4,04d5,04d6,04d7,04d8,04d9,04da,04db,04dc,04dd,04de,04df,04e0,04e1,04e2,04e3,04e4,04e5,04e6,04e7,04e8,04e9,04ea,04eb,04ec,04ed,04ee,04ef,04f0,04f1,04f2,04f3,04f4,04f5,04f6,04f7,04f8,04f9,04fa,04fb,04fc,04fd,04fe,04ff,0500,0501,0502,0503,0504,0505,0506,0507,0508,0509,050a,050b,050c,050d,050e,050f,0510,0511,0512,0513,0514,0515,0516,0517,0518,0519,051a,051b,051c,051d,051e,051f,0520,0521,0522,0523,0524,0525,0526,0527,0528,0529,052a,052b,052c,052d,052e,052f,0530,0531,0532,0533,0534,0535,0536,0537,0538,0539,053a,053b,053c,053d,053e,053f,0540,0541,0542,0543,0544,0545,0546,0547,0548,0549,054a,054b,054c,054d,054e,054f,0550,0551,0552,0553,0554,0555,0556,0557,0558,0559,055a,055b,055c,055d,055e,055f,0560,0561,0562,0563,0564,0565,0566,0567,0568,0569,056a,056b,056c,056d,056e,056f,0570,0571,0572,0573,0574,0575,0576,0577,0578,0579,057a,057b,057c,057d,057e,057f,0580,0581,0582,0583,0584,0585,0586,0587,0588,0589,058a,058b,058c,058d,058e,058f,0590,0591,0592,0593,0594,0595,0596,0597,0598,0599,059a,059b,059c,059d,059e,059f,05a0,05a1,05a2,05a3,05a4,05a5,05a6,05a7,05a8,05a9,05aa,05ab,05ac,05ad,05ae,05af,05b0,05b1,05b2,05b3,05b4,05b5,05b6,05b7,05b8,05b9,05ba,05bb,05bc,05bd,05be,05bf,05c0,05c1,05c2,05c3,05c4,05c5,05c6,05c7,05c8,05c9,05ca,05cb,05cc,05cd,05ce,05cf,05d0,05d1,05d2,05d3,05d4,05d5,05d6,05d7,05d8,05d9,05da,05db,05dc,05dd,05de,05df,05e0,05e1,05e2,05e3,05e4,05e5,05e6,05e7,05e8,05e9,05ea,05eb,05ec,05ed,05ee,05ef,05f0,05f1,05f2,05f3,05f4,05f5,05f6,05f7,05f8,05f9,05fa,05fb,05fc,05fd,05fe,05ff,0600,0601,0602,0603,0604,0605,0606,0607,0608,0609,060a,060b,060c,060d,060e,060f,0610,0611,0612,0613,0614,0615,0616,0617,0618,0619,061a,061b,061c,061d,061e,061f,0620,0621,0622,0623,0624,0625,0626,0627,0628,0629,062a,062b,062c,062d,062e,062f,0630,0631,0632,0633,0634,0635,0636,0637,0638,0639,063a,063b,063c,063d,063e,063f,0640,0641,0642,0643,0644,0645,0646,0647,0648,0649,064a,064b,064c,064d,064e,064f,0650,0651,0652,0653,0654,0655,0656,0657,0658,0659,065a,065b,065c,065d,065e,065f,0660,0661,0662,0663,0664,0665,0666,0667,0668,0669,066a,066b,066c,066d,066e,066f,0670,0671,0672,0673,0674,0675,0676,0677,0678,0679,067a,067b,067c,067d,067e,067f,0680,0681,0682,0683,0684,0685,0686,0687,0688,0689,068a,068b,068c,068d,068e,068f,0690,0691,0692,0693,0694,0695,0696,0697,0698,0699,069a,069b,069c,069d,069e,069f,06a0,06a1,06a2,06a3,06a4,06a5,06a6,06a7,06a8,06a9,06aa,06ab,06ac,06ad,06ae,06af,06b0,06b1,06b2,06b3,06b4,06b5,06b6,06b7,06b8,06b9,06ba,06bb,06bc,06bd,06be,06bf,06c0,06c1,06c2,06c3,06c4,06c5,06c6,06c7,06c8,06c9,06ca,06cb,06cc,06cd,06ce,06cf,06d0,06d1,06d2,06d3,06d4,06d5,06d6,06d7,06d8,06d9,06da,06db,06dc,06dd,06de,06df,06e0,06e1,06e2,06e3,06e4,06e5,06e6,06e7,06e8,06e9,06ea,06eb,06ec,06ed,06ee,06ef,06f0,06f1,06f2,06f3,06f4,06f5,06f6,06f7,06f8,06f9,06fa,06fb,06fc,06fd,06fe,06ff,0700,0701,0702,0703,0704,0705,0706,0707,0708,0709,070a,070b,070c,070d,070e,070f,0710,0711,0712,0713,0714,0715,0716,0717,0718,0719,071a,071b,071c,071d,071e,071f,0720,0721,0722,0723,0724,0725,0726,0727,0728,0729,072a,072b,072c,072d,072e,072f,0730,0731,0732,0733,0734,0735,0736,0737,0738,0739,073a,073b,073c,073d,073e,073f,0740,0741,0742,0743,0744,0745,0746,0747,0748,0749,074a,074b,074c,074d,074e,074f,0750,0751,0752,0753,0754,0755,0756,0757,0758,0759,075a,075b,075c,075d,075e,075f,0760,0761,0762,0763,0764,0765,0766,0767,0768,0769,076a,076b,076c,076d,076e,076f,0770,0771,0772,0773,0774,0775,0776,0777,0778,0779,077a,077b,077c,077d,077e,077f,0780,0781,0782,0783,0784,0785,0786,0787,0788,0789,078a,078b,078c,078d,078e,078f,0790,0791,0792,0793,0794,0795,0796,0797,0798,0799,079a,079b,079c,079d,079e,079f,07a0,07a1,07a2,07a3,07a4,07a5,07a6,07a7,07a8,07a9,07aa,07ab,07ac,07ad,07ae,07af,07b0,07b1,07b2,07b3,07b4,07b5,07b6,07b7,07b8,07b9,07ba,07bb,07bc,07bd,07be,07bf,07c0,07c1,07c2,07c3,07c4,07c5,07c6,07c7,07c8,07c9,07ca,07cb,07cc,07cd,07ce,07cf,07d0,07d1,07d2,07d3,07d4,07d5,07d6,07d7,07d8,07d9,07da,07db,07dc,07dd,07de,07df,07e0,07e1,07e2,07e3,07e4,07e5,07e6,07e7,07e8,07e9,07ea,07eb,07ec,07ed,07ee,07ef,07f0,07f1,07f2,07f3,07f4,07f5,07f6,07f7,07f8,07f9,07fa,07fb,07fc,07fd,07fe,07ff,0800,0801,0802,0803,0804,0805,0806,0807,0808,0809,080a,080b,080c,080d,080e,080f,0810,0811,0812,0813,0814,0815,0816,0817,0818,0819,081a,081b,081c,081d,081e,081f,0820,0821,0822,0823,0824,0825,0826,0827,0828,0829,082a,082b,082c,082d,082e,082f,0830,0831,0832,0833,0834,0835,0836,0837,0838,0839,083a,083b,083c,083d,083e,083f,0840,0841,0842,0843,0844,0845,0846,0847,0848,0849,084a,084b,084c,084d,084e,084f,0850,0851,0852,0853,0854,0855,0856,0857,0858,0859,085a,085b,085c,085d,085e,085f,0860,0861,0862,0863,0864,0865,0866,0867,0868,0869,086a,086b,086c,086d,086e,086f,0870,0871,0872,0873,0874,0875,0876,0877,0878,0879,087a,087b,087c,087d,087e,087f,0880,0881,0882,0883,0884,0885,0886,0887,0888,0889,088a,088b,088c,088d,088e,088f,0890,0891,0892,0893,0894,0895,0896,0897,0898,0899,089a,089b,089c,089d,089e,089f,08a0,08a1,08a2,08a3,08a4,08a5,08a6,08a7,08a8,08a9,08aa,08ab,08ac,08ad,08ae,08af,08b0,08b1,08b2,08b3,08b4,08b5,08b6,08b7,08b8,08b9,08ba,08bb,08bc,08bd,08be,08bf,08c0,08c1,08c2,08c3,08c4,08c5,08c6,08c7,08c8,08c9,08ca,08cb,08cc,08cd,08ce,08cf,08d0,08d1,08d2,08d3,08d4,08d5,08d6,08d7,08d8,08d9,08da,08db,08dc,08dd,08de,08df,08e0,08e1,08e2,08e3,08e4,08e5,08e6,08e7,08e8,08e9,08ea,08eb,08ec,08ed,08ee,08ef,08f0,08f1,08f2,08f3,08f4,08f5,08f6,08f7,08f8,08f9,08fa,08fb,08fc,08fd,08fe,08ff,0900,0901,0902,0903,0904,0905,0906,0907,0908,0909,090a,090b,090c,090d,090e,090f,0910,0911,0912,0913,0914,0915,0916,0917,0918,0919,091a,091b,091c,091d,091e,091f,0920,0921,0922,0923,0924,0925,0926,0927,0928,0929,092a,092b,092c,092d,092e,092f,0930,0931,0932,0933,0934,0935,0936,0937,0938,0939,093a,093b,093c,093d,093e,093f,0940,0941,0942,0943,0944,0945,0946,0947,0948,0949,094a,094b,094c,094d,094e,094f,0950,0951,0952,0953,0954
```

Encrypted Client Hello

- ECH extension
- Plain data are bogus
- You first need to decrypt
- This kills PCAPs



<https://blog.miloslavhomer.cz/encrypted-client-hello/>

JA3 Algorithm

```
def ja3(...):  
    t = str(tls_version)  
    c = "-".join([str(x) for x in ciphers])  
    ex = "-".join([str(x.id) for x in extensions])  
    ec = "-".join([str(x) for x in elliptic_curves])  
    pf = "-".join([str(x) for x in elliptic_curve_point_formats])  
    in_str = f"{t},{c},{ex},{ec},{pf}"  
    return hashlib.md5(in_str.encode()).hexdigest()
```

TLS Fingerprinting Timeline

2015 - TLS Fingerprint idea ([blog](#))

2017 - ja3 ([github](#))

2020 - GREASE [RFC8701](#)

2022 - Chrome extension permutation [feature](#)

2023 - JA4 ([blog](#))

GREASE (RFC8701)

Generate Random Extensions And Sustain Extensibility

0x0A0A, 0x1A1A, 0x2A2A, 0x3A3A,

0x4A4A, 0x5A5A, 0x6A6A, 0x7A7A,

0x8A8A, 0x9A9A, 0xAAAA, 0xBABA,

0xCACA, 0xDADA, 0xEAEA, 0xFAFA,

JA4+

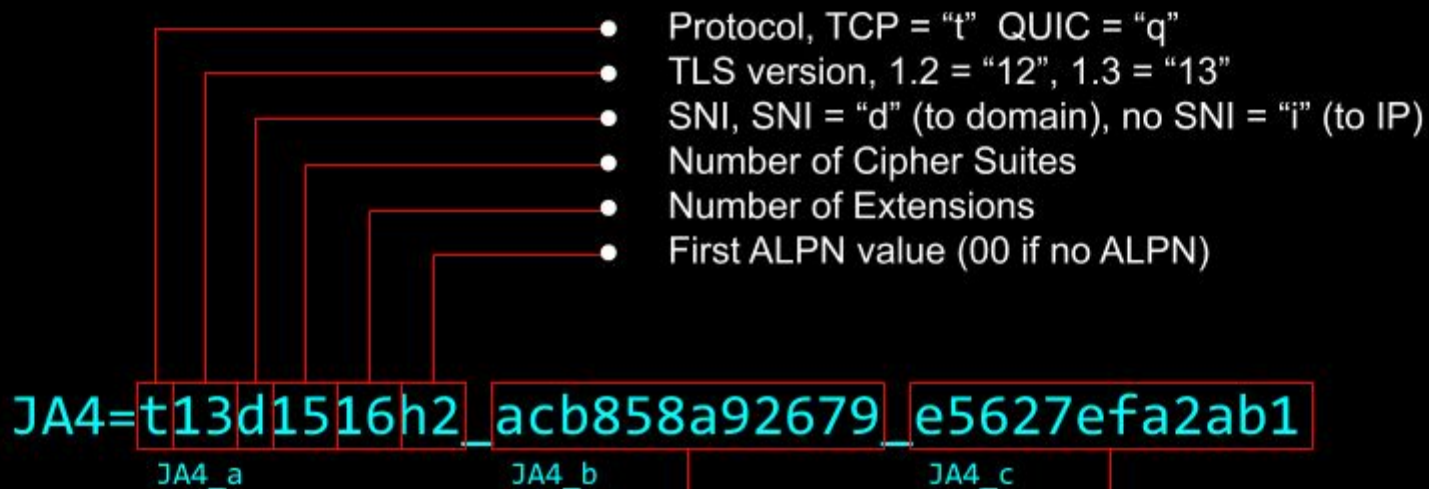
JA4+

- Suite of Hashes
 - Server Side
 - HTTP client
 - SSH
- John Althouse
- Jeff Atkinson
- Josh Atkins
- Joshua Alexander
- And many others!



<https://github.com/FoxIO-LLC/ja4>

JA4: TLS Client Fingerprint



- Truncated SHA256 hash of the Cipher Suites, sorted
- Truncated SHA256 hash of the Extensions, sorted
+ Signature Algorithms, in the order they appear

TLS Fingerprinting Timeline

2015 - TLS Fingerprint idea ([blog](#))

2017 - ja3 ([github](#))

2020 - GREASE [RFC8701](#)

2022 - Chrome extension permutation ([feature](#))

2023 - JA4 ([blog](#))

2024 - Cloudflare implements JA4 ([blog](#))

Deploying JA4+

- ✓ Big Providers
- ✓ Open Source / Self-hosted
 - Proxies: HAProxy, Nginx
 - IDS/IPS: Suricata, Zeek, ...
 - PCAP: Wireshark, Caploader, Arkime ...
 - Go Libs - X509 only



<https://github.com/FoxIO-LLC/ja4?tab=readme-ov-file#tools-that-support-ja4>

JA4+ Licensing

- I AM NOT A LAWYER
- JA4: TLS Client Fingerprinting: BSD 3-Clause
- Rest of JA4+: FoxIO License 1.1
 - Academic, internal business use good
 - Monetization bad
 - You can secure your own assets, you can't sell protection

JA4DB: Gone?



JA4+ Database

Download, read, learn about, and contribute to augment your organization's JA4+ network security efforts

Download the database

Read the database



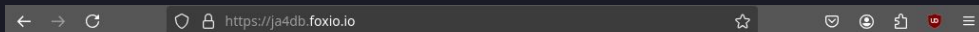
Download and read the database to round out your organization's JA4+ defenses



Learn about the fingerprints in the JA4+ suite and which applications, certificate authorities, libraries, and operating systems they're associated with



Contribute to the database to strengthen network security for yours and other organizations



Welcome

Access the FoxIO database and learn more about network traffic you are experiencing

- Secure authentication
- Advanced encryption

Sign In

Enter your credentials to access your account

Email Address

Enter your email

Password

Enter your password

☐ Remember me

[Forgot password?](#)

Sign In

Don't have an account?

[Sign up](#)

JA4 in Practice

Deploying JA4 on HAProxy

```
global
```

```
    lua-load /etc/haproxy/lua/ja4.lua  
    tune.ssl.capture-buffer-size 2048
```

```
frontend https-in
```

```
    http-request lua.fingerprint_ja4  
    http-request set-header X-JA4 var(txn.fingerprint_ja4)  
    http-request capture var(txn.fingerprint_ja4) len 36  
    http-request capture req.hdr(user-agent) len 150
```



<https://blog.miloslavhomer.cz/deploying-ja4/>

Blog Analysis

- Packets into PCAPs
- Compute ja4 hashes
- Check

ChatGPT-User

- Mozilla/5.0 AppleWebKit/537.36 (KHTML, like Gecko); compatible; ChatGPT-User/1.0; +https://openai.com/bot
- t13d1711h2_5b57614c22b0_78e6aca7449b
- t13d1712h2_5b57614c22b0_d5fe2c511efa
- t13d1712h2_5b57614c22b0_ef7df7f74e48
- t13d1713h2_5b57614c22b0_02c8e53ee398
- Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/131.0.0.0 Safari/537.36; compatible; OAI-SearchBot/1.3; robots.txt; +https://openai.com/searchbot

ChatGPT-User

- Mozilla/5.0 AppleWebKit/537.36 (KHTML, like Gecko); compatible; ChatGPT-User/1.0; +https://openai.com/bot
- t13d1711h2_5b57614c22b0_78e6aca7449b
- t13d1712h2_5b57614c22b0_d5fe2c511efa
- t13d1712h2_5b57614c22b0_ef7df7f74e48
- t13d1713h2_5b57614c22b0_02c8e53ee398
- Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/131.0.0.0 Safari/537.36; compatible; OAI-SearchBot/1.3; robots.txt; +https://openai.com/searchbot

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- t13d1712h2_5b57614c22b0_d5fe2c511efa
- t13d1712h2_5b57614c22b0_ef7df7f74e48
- t13d1713h2_5b57614c22b0_02c8e53ee398
- Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/131.0.0.0 Safari/537.36; compatible; OAI-SearchBot/1.3; robots.txt; +https://openai.com/searchbot

ChatGPT-User

- t13d1711h2_5b57614c22b0_5894756fee65
- t13d1711h2_5b57614c22b0_78e6aca7449b
- t13d1712h2_5b57614c22b0_d5fe2c511efa
- t13d1712h2_5b57614c22b0_ef7df7f74e48
- t13d1713h2_5b57614c22b0_02c8e53ee398

ChatGPT-User

- t13d1711h2_5b57614c22b0_5894756fee65
- Mozilla/5.0 AppleWebKit/537.36 (KHTML, like Gecko); compatible; ShapBot/0.1.0
- 136.119.129.236
- 23.236.55.230
- 34.173.239.49
- 34.56.168.230
- 34.60.131.171
- <https://docs.parallel.ai/resources/shapbot.json>

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2026 - Encrypted Client Hello [RFC9849](#)

Updated JA4+?



John Althouse ✓ • 1st
Founder and CTO at FoxIO
4mo • Edited •

...

We've stopped doing blog posts on a regular basis.

Over the past 2 years we've developed many new technologies and capabilities that now solve most cyber security challenges but we're not talking about them publicly. Why? Because what used to take actors years, we're now seeing them pivot within days. So instead, we're presenting these findings and capabilities directly to practitioners at conferences and by providing consulting services and technologies to some of the largest companies in the world, to help them solve their toughest challenges, directly.

You will still see blog posts from us when we publish new JA4+ methods but the old style of publicly blogging about flipping the table on actors, dropping the mic and moonwalking off stage is over. We're still doing it, just quietly with our customers and industry practitioners.



What About Attackers?

Client Has Full Control Over ClientHello

That means they can mimic:

- <https://github.com/lwthiker/curl-impersonate>
- <https://github.com/enetx/surf>
- <https://github.com/refraction-networking/utls>
- <https://github.com/0x676e67/wreq>

But plenty of clients are lazy.

Conclusion

JA4+ is not a silver bullet

- The clients can update
 - The attackers can mimic
 - It's not 100% consistent
 - Hash DB???
-
- But when it hits

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 - The attackers can mimic
 - It's not 100% consistent
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- But when it hits



Thank you! Questions?



<https://www.linkedin.com/in/miloslav-homer/>

